Rahul Deopa

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ACADEMIC DETAILS

COURSE	SPECIALIZATION	INSTIUTE/ COLLEGE	BOARD/ UNIVERSITY	SCORE	YEAR
PhD	Water Resources Development and Management	Indian Institute of Technology Roorkee	Indian Institute of Technology Roorkee	9.33 (CPI)	2023-Present
PG	Water Resources Engineering (Civil Engineering)	National Institute of Technology, Kurukshetra	NIT Kurukshetra	9.34 (CGPA)	2022
UG	Civil Engineering	Seemant Institute of Technology	Uttarakhand Technical University	73.36 %	2018

RESEARCH INTEREST

- **Flood risk management:** Hydrodynamic flood inundation and hazard mapping, Urban flooding, Human health risk, Water quality, multi-hazard flood risk assessment
- **Analysis of hydro-climatic extremes:** Regionalisation and design rainfall analysis for data-poor catchments, Statistical downscaling of extreme rainfall events, Climate Change

PUBLICATIONS

- **Deopa, R.**, Tripathi, V., Nandana, P. S., & Mohanty, M. P. (2026). Coupled hydrological and public health risks from urban flooding: Integrated remote sensing, machine learning, and hydrodynamic–ecological modelling. ***Journal of Hydrology***, Article 135999. https://doi.org/10.1016/j.jhydrol.2026.135999
- **Deopa, R.,** Singh, H., Tripathi, V., Tyagi, M., & Mohanty, M. P. (2025). Cascading impacts of 2025 June-July multi-hazard episodes in the Western Himalayas: Evidence through remote sensing observations and Google Earth Engine. ***Natural Hazards (2025)***. https://doi.org/10.1007/s11069-025-07699-x.
- **Deopa, R.**, & Mohanty, M. P. (2025). HyEco: A Hybrid Hydrodynamic-cum-Ecological Modelling Framework to Demystify Flood and Human-Health Risks over Flood-prone Metropolis. ***Water Research X***, 100396. https://doi.org/10.1016/j.wroa.2025.100396
- **Deopa, R.**, & Mohanty, M. P. (2025). HyEco-R: Integrated Hydrodynamic & Water Quality Simulations for Flood-Prone Urban Metropolis [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.16270304
- **Deopa, R.**, Thakur, D. A., Kumar, S., Mohanty, M. P., & Asha, P. (2024). Discerning the dynamics of urbanization-climate change-flood risk nexus in densely populated urban mega cities: An appraisal of efficient flood management through spatiotemporal and geostatistical rainfall analysis and hydrodynamic modeling. ***Science of The Total Environment***, 175882. https://doi.org/10.1016/j.scitotenv.2024.175882

CONFERENCES

- **Deopa, R.**, Mishra, D., Dayal, D., Singh, H., & Mohanty, M. P. (2026, May 27). Integrated modelling of the flood–hydro-ecological–health nexus under future climate extremes in urban megacities. **JpGU–AGU Joint Meeting 2026, Chiba, Japan**.

- **Deopa, R.,** Mishra, D., Dayal, D., Shahi, N. K., and Mohanty, M. P.: Hydrology–Health Nexus in a Changing Climate: Multi-Hazard Modelling of Cascading Flood–Health Risks in Urban Megacities, **EGU General Assembly 2026**, Vienna, Austria, 3–8 May 2026, EGU26-328, https://doi.org/10.5194/egusphere-egu26-328, 2026.

- Singh, P., **Deopa, R.**, and Mohanty, M. P.: Assessing urban surface flood resilience using hydrodynamic modelling under extreme rainfall conditions in urban catchment of Nepal, **EGU General Assembly 2026, Vienna, Austria**, 3–8 May 2026, EGU26-18481, https://doi.org/10.5194/egusphere-egu26-18481, 2026.

- Saha, S., **Deopa, R**., Pandey, A., Rao, S., & Mohanty, M. P. (2025). Integrated Web-Based Framework for Risk Simulation and Visualization of Potentially Dangerous Glacial Lakes in The Indian Himalayan Region. AGU25.

- **Deopa, R.**, & Mohanty. M. P. (2025)**.** HyEco: An Integrated Hydrodynamic–Ecological Model to Quantify Human Health Risks from Contaminated Floodwaters in Climate-Sensitive Urban Megacities. **XIIth IAHS Scientific Assembly 2025**. **October 5-10, 2025**

- Mohanty, M.P., & **Deopa, R.** (2025)**.** HyEcO: An integrated framework to bridge hydrodynamic and ecological risks during unprecedented climate extremes. **ACS General Conference**. **August 19, 2025**.

- **Deopa, R.,** Mishra, D., Kumar Shahi, N., Tripathi, V., & Prakash Mohanty, M.: Quantifying human health risks from contaminated urban floodwaters using a multi-model framework under climate and socio-economic scenarios. **EGU General Assembly 2025**, Vienna, Austria, **27 Apr–2 May 2025**, EGU25-15444, https://doi.org/10.5194/egusphere-egu25-15444, 2025.

- **Deopa, R.**, & Mohanty. M. P. (2024). An integrated framework to bridge and map flood risks and human health risks from contaminated floodwaters during urban flooding episodes. **9th International Symposium on Integrated Water Resource Management (IWRM)**, **14th International Workshop on Statistical Hydrology (STAHY)**. **Nov. 04-07, 2024**, Florianopolis, Brazil.

- **Deopa, R.**, & Mohanty. M. P. (2024). Quantifying the Impact of Urbanization and Rising Rainfall Extremes on Flood Risks in Delhi through Spatio-Temporal Analysis and Hydrodynamic Modeling. **Young Leaders Conclave (YLC) 2024**, **Jun. 04-05, 2024**, Delhi, India.

- **Deopa, R.**, Asha, P., & Mohanty. M. P. (2024). Assessing the Impacts of Diverse Land Use Patterns on Water Quality and Human Health Risks Associated with Contaminated Floodwaters**. ICFWR 2023**, **Jan. 18-20, 2024**, Roorkee, India.

- **Deopa, R.,** Tripathi, V., & Mohanty. M. P. (2023). Capturing the teleconnections of climate change and socio-economic structures on flood and human health risks from contaminated urban flood. **WCRP Open Science Conference 2023**, **Oct. 23-27, 2023**, Rwanda.

- **Deopa, R.,** & Mohanty. M. P. (2023). Integrating regional climate models in flood modeling to quantify human health risks from contaminated urban floodwaters, **ICRC-CORDEX 2023, Sep. 25-29, 2023**. Trieste, Italy. *(Received travel grant for presentation)*

BOOK CHAPTERS

- **Deopa, R.**, Singh, K.K. (2024). Influence on Water Characteristics Away from Various Sources of NIT Kurukshetra Using ArcGIS. In: Mesapam, S., Ohri, A., Sridhar, V., Tripathi, N.K. (eds) Developments and Applications of Geomatics. DEVA 2022. Lecture Notes in Civil Engineering, vol 450. Springer, Singapore. https://doi.org/10.1007/978-981-99-8568-5_24

- Deopa, R., Singh. K.K. (2022). Assessment of Bleaching Powder Performance of Drinking Water Supply Network of NIT Campus using GIS based Mike Plus, **International Virtual Conference on Developments and Applications of Geomatics (DEVA-2022),** Page- 106, NIT Warangal

PROJECTS/TECHNICAL REPORTS

- Mohanty, M.P., & **Deopa, R.** (2025). "*An integrated approach to quantify the impacts on climate change and socioeconomic dynamics on human health risk associated with the contaminated floodwaters*", funded by the **Anusandhan National Research Foundation (ANRF)**, **Department of Science and Technology (DST), Government of India (Project No. SRG/2022/001094)**

  (*Nov 2023 to Jan 2025)*

- Completed a technical project titled "*Assessment of Water Characteristics of the Drinking Water Supply Network of NIT Kurukshetra Campus using MIKE+ Software*" involving hydraulic and water quality modeling of the campus distribution system to evaluate flow dynamics, pressure zones, and residual disinfectant levels for performance optimization and infrastructure planning. (***November 2020 to June 2023)***

- Completed the interactive track and course project of "**Computational Tools for Climate Science – 2024/2025**" offered by Climate Match Academy, comprising 85 hours of training in climate data analysis, modeling workflows, and coding-based applications for climate science research. ***(July 2024, 2025)***

WORKSHOPS/TRAININGS/LECTURES

Courses

- Teaching Assistant for the course on "Drinking-Water and Sanitation Sustainability (WRN-507)" for the Autum Semester 2024-25 at Department of WRD&M, IIT Roorkee.

Trainings conducted/attended

- **Conducted 2-months training program** on "Water, built environment, and emergency preparedness" under the aegis of the National Health Mission for Odisha State engineers (March-May 2024)
- **Conducted a Hands-on session on "**Developing an exhaustive 1D-2D coupled flood model set-up using MIKE+" during Recent Advances in Satellite-based Technologies for monitoring Hydro-climatological Extremes (RASTER - 2023), Organized by the Department of Water Resources Development and Management, Indian Institute of Technology Roorkee on 13th September 2023
- Participated program one-week basic course on "Urban Flood Modelling in a Changing Climate" from June 06-12, 2024, held at BITS Pilani-Hyderabad Campus
- Participated in one-day workshop on "Water Quality and its Interlinkages with Sustainable Development" - Dept. of WRDM, IIT Roorkee (May 2023)
- Participated in Departmental Research Scholar Day-2023 (**Jal Utkarsh) –** Dept. of WRDM, IIT Roorkee (April 2023)

Special Lectures

- **Conducted professional training** for L&T engineers on "Flood Risk Analysis Program" bridging the gap between advanced hydrological modeling and industrial application at Chennai Headquarters (16-18 October 2025)

AWARDS & ACHIEVEMENTS

- **Awarded travel grant** to attend a one-week basic course on "Urban Flood Modelling in a Changing Climate" from June 06-12, 2024, held at BITS Pilani-Hyderabad Campus
- Awarded **Best Paper** Award at the **Young Leaders Conclave 2024** by National Disaster Management Authority (NDMA), New Delhi, India.
- Awarded **Travel Grant** by **World Meteorological Organisation (WMO)** & **UNESCO** for presentation in the International Conference on Regional Climate-CORDEX 2023, Sep. 25-29, 2023, Trieste, Italy.
- Awarded **Junior Research Fellowship** at IIT Roorkee under DST-sponsored project titled" An integrated approach to quantify the human health risk associated with the urban floods".
- Awarded **MHRD fellowship** by Govt. of India to pursue M. Tech degree in Water Resources Engineering at NIT Kurukshetra, Haryana, India. **(July 2020 to July 2022)**
- Qualified **Graduate Aptitude Test in Engineering (GATE) 2019** and **2020** scholarship in Civil Engineering for masters.

SOFT SKILLS PROGRAMS

CERTIFICATIONS	***CERTIFYING AUTHORITY***
Computational Tools for Climate Science – 2024/2025	***Climatematch Academy***
MIKE FLOOD – Getting started with flood modelling	***Danish Hydraulic Institute (DHI)***
MIKE ECO Lab – Creating a framework for water quality and ecological modelling	***Danish Hydraulic Institute (DHI)***
Modelling of Water Supply System Certificates	***Danish Hydraulic Institute (DHI)***
MIKE + Addressing Water Distribution Model Holistically	***Danish Hydraulic Institute (DHI)***
Certificate of Accomplishment on "Big Data Applications in Water Resources and Hydro Informatics"	***Ministry of Jal Shakti (Govt. of India)***
Certificate of Accomplishment on "Introduction to Python and its applications to Water Resources Sector"	***Ministry of Jal Shakti (Govt. of India)***

EXTRACURRICULAR

- Member of Institute Cricket Squad and represented IIT Roorkee at the **58th Inter-IIT Sports Meet 2024,** IIT Hyderabad ***December 2025***

- Won **Gold Medal** in Cricket Tournament in Sangram Annual Sports Fest, IIT Roorkee — ***September 2025***
- Served as a Bhawan and Cultural ***Council Member*** in the Hostel (Vigyan Bhawan) at IIT Roorkee — ***August 2024-November2025***
- Participated in multiple sports events during **the Inter-Bhawan Sports Tournament 2024–25**, IIT Roorkee — ***March-April 2025***
- Selected for the Institute Cricket Squad and represented IIT Roorkee at the **57th Inter-IIT Sports Meet 2024,** Kanpur. — ***December 2024***
- Participated in the Annual Sports Fest "Sangram 2024" – IIT Roorkee — ***March 2024***
- Committee Member at Post Graduate Club at National Institute of Technology, Kurukshetra — ***Feb 2022 - June 2022***
- Event Coordinator at ABACUS event of College Fest – Seemant Institute of Technology (Uttarakhand Technical University) — ***April 2018***
- Student Editor of College Annual Magazine – Seemant Institute of Technology (UTU) — ***Sep 2015 – Sep 2016***
- Organizer for Cricket Event in Annual Sports Meet GARJANA - Seemant Institute of Technology (UTU) — ***September 2016***

ADDITIONAL RESPONSIBILITIES

- **Part of the organizing team** during the Roorkee Water Conclave (RWC) 2026 organized by Indian Institute of Technology Roorkee and National Institute of Hydrology (23-25 February 2026)
- **Part of the organizing team** during the International Conference on Future of Water Resources (ICFWR-2024) organized by Indian Water Resources Society and Department of Water Resources Development and Management (WRD&M), IIT Roorkee (18-20 January 2024)
- **Multiple field visits** in Yamuna floodplains in Delhi during the 2023 Delhi flood events to analyse the situation and collection of the field data.
- **Part of the organizing team** during 7-day workshop on "Recent Advancers in Satellite based Technologies for monitoring hydroclimatological ExtRemes (RASTER-2023) organized by the Department of Water Resources Development and Management (WRD&M), IIT Roorkee (September 2023)
- **Captain** of Hostel Cricket Team during Inter-Bhawan Sports Meet organized by Institute Sports Council (ISC) at Indian Institute of Technology Roorkee (February 2025)

I, the undersigned, certify that to the best of my knowledge and belief, this personnel biodata/CV correctly describes myself, my qualifications, and my experience.

RAHUL DEOPA
26-02-2026
Signature

Google Scholar

LinkedIn